

DIVYANG SINGH SOMVANSHI

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SUMMARY

Computer Science Engineer focused on Artificial Intelligence, Machine Learning and Data Analytics. Expert in fundamentals of **Python, Data Structures and Algorithms**, with practical knowledge of developing agentic workflows using LLMs, RAG pipeline development, and RESTful API development based on well-designed databases and data pipelines. Analytical, accountable, and passionate professional turning real-world data into reliable, production-ready systems while collaborating openly across platforms.

TECHNICAL SKILLS

Programming Language: Python, C++, SQL

Machine Learning: Scikit-learn, Tensorflow, Feature Engineering, MLflow, Cross-Validation, Model Training, Model Evaluation

Artificial Intelligence: PyTorch, Hugging Face, Transformer, Retrieval-Augmented Generation (RAG) Pipelines, LangChain, Model Context Protocol (MCP), Agentic Workflows, Prompt Engineering

Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn, Power BI

APIs & Backend Development: FastAPI, LLM APIs, FAISS (Vector Database), Docker

Cloud Computing & Tools: AWS, GitHub, Visual Studio Code, Streamlit, Jupyter Notebook

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP)

PROJECTS

AI Research Paper Analysis Tool | [GitHub Link](#) | *Python, NLP, LLM, LangChain, PyTorch, FastAPI*

- Designed a RAG pipeline using FAISS vector database and Sentence Transformers that gave around 92% semantic similarity retrieval accuracy on 150+ research papers.
- Designed a full-stack AI/ML data pipeline for (PDF parsing, chunking, indexing, embedding, LLM-based Q&A) using LangChain, Hugging Face Transformers, PyTorch, deployed as a modular FastAPI backend (RESTful APIs) serving 10+ queries concurrently with <2s latency and reducing manual review by about 60%.
- Implemented an LLM-based agentic summary pipeline for automatically generating structured research summaries that reduced document comprehension time from 45 min per paper to less than 5 min per paper.

Smart Study Chat App | [GitHub Link](#) | *Python, Streamlit, OpenAI API, LangChain*

- Developed a multi-PDF AI Question & Answering system using Python, LangChain agents, and the OpenAI API, deployed on Streamlit Cloud, capable of simultaneously uploading over 20 documents.
- Semantic Similarity Search was implemented using FAISS and Sentence Transformers to deliver more than 85% accurate context-matching on 1500+ queries in real-life messy data. The tokenization process was optimized and reduced latency time from 4 seconds to 1.2 seconds.
- Improved Hugging Face embeddings and iterative training through user feedback loop increased the relevance of answers by about 30% comparing to keyword-based searches.

Stock Market Portfolio Tracker | [GitHub Link](#) | *Python, Pandas, SQL, Scikit-learn, Matplotlib*

- Developed a portfolio analytics platform in Python using core data structures & algorithms, Pandas, and SQL to track and visualize performance of 15+ stocks with real-time trend dashboards.
- Developed Linear Regression and Random Forest models for predicting stock price trends with a mean absolute error (MAE) of 3.2% based on 6 months of back-tested data. Developed different technical indicators (RSI, MACD), improving efficiency by about 22%.
- Added real-time signals from the market by developing news sentiment analysis (VADER + BeautifulSoup), improving decision-making process efficiency by about 18%.

EDUCATION

Bachelor of Technology, Computer Science & Engineering

SRM Institute of Science and Technology, Ghaziabad, India

September 2022 – May 2026

CGPA: **8.0 / 10.0**

St. Xavier's Senior Secondary School (XII)

Central Board of Secondary Education

2021-2022

Percentage: **68.83%**

St. Xavier's Senior Secondary School (X)

Central Board of Secondary Education

2019-2020

Percentage: **89%**

CERTIFICATIONS

- OCI AI Foundations Associate - Oracle University
- Machine Learning - Coursera
- Database Management System - NPTEL

EXTRACURRICULAR & LEADERSHIP

INNOVATE 2024 | *Public Relations Member* | *National Level Technical Fest*

April 2024 – April 2024

- Held position of Public Relations member in a national technology festival that comprised activities such as RoboWars. Took responsibility of branding the event, conducted campaigns and developed partnerships for workshops and exhibits.

RUBAROO CULTURAL FEST | *Management Volunteer* | *National Level Cultural Fest*

April 2023 – April 2023

- Provided coordination support for a nation-wide cultural festival that had over 5,000 students attend it and provided logistics planning assistance, artist management, crowd control support to make sure the performance was executed efficiently.